

Module 8: Data Validation, Dropdowns & Conditional Formatting

Duration: 32 minutes | **Learn:** 10 min | **Lab:** 22 min

Control what goes **IN** to your spreadsheet. Make what's already there **VISIBLE** at a glance. According to a 2023 Gartner report, poor data quality costs organisations an average of **\$12.9 million per year** — validation and formatting are your first line of defence.

DEVELOPED BY TALENCIAGLOBAL

EXCEL MASTERY SERIES

Data Validation: Controlling Input

Data Validation restricts what users can type into a cell — preventing errors before they happen. Studies show that **88% of spreadsheets contain at least one error** (University of Hawaii, Raymond Panko), the majority of which stem from unconstrained manual data entry.

 **Where to find it:** Data tab → Data Validation

Validation Types

Allow	What It Checks	Example Use
Whole Number	Must be an integer within a range	Age: between 18 and 65
Decimal	Must be a decimal within a range	Percentage: between 0.0 and 100.0
List	Must be one of the predefined options	Department: from a dropdown list
Date	Must be a date within a range	Join Date: between 1-Jan-2015 and today
Time	Must be a time within a range	Shift start: between 08:00 and 20:00
Text Length	Text must be a certain length	Employee ID: exactly 7 characters
Custom	Must satisfy a custom formula	No duplicates: =COUNTIF(\$A:\$A, A2)<=1

Three Tabs in the Data Validation Dialog

1

Settings
Define the rule — type, operator, and values. This is where the constraint logic lives.

2

Input Message
A tooltip that appears when the cell is selected. Example: "Please select a department."

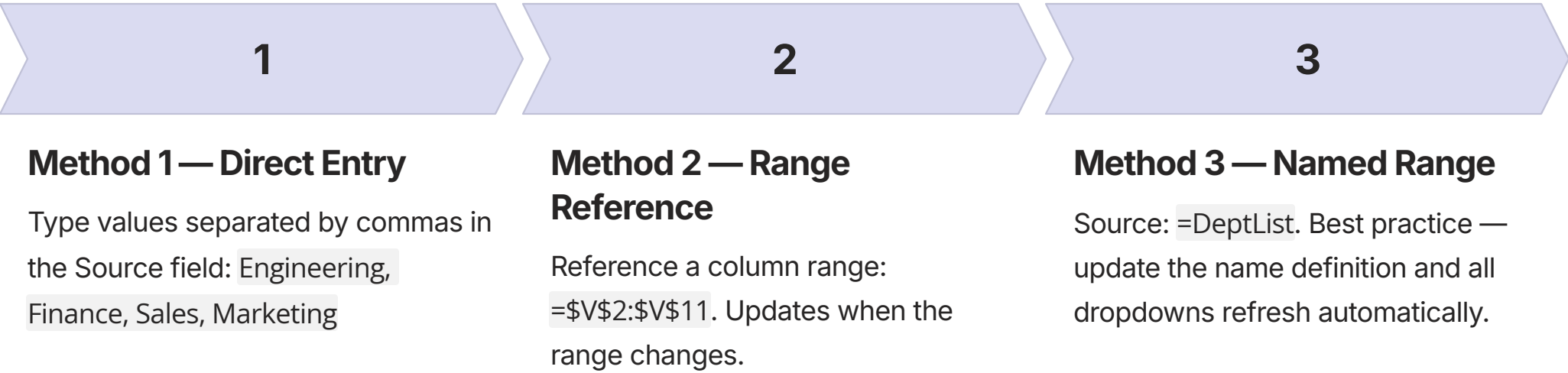
3

Error Alert
What happens when invalid data is entered — Stop, Warning, or Information.

 **Demo:** Apply whole-number validation on the Experience_Years column — allow only integers between 0 and 50. Set an Input Message: "Enter years of experience (0–50)." Set an Error Alert: Stop → "Invalid entry. Experience must be between 0 and 50." Then type 60 to trigger the error.

Dropdown Lists

Dropdowns are the most commonly used validation type — they present a predefined list of choices, eliminating free-text entry errors entirely. In enterprise HR systems, dropdown-enforced fields reduce data cleansing time by up to **60%** compared to open-text fields.



Dependent / Cascading Dropdowns

The second dropdown's options change based on the first dropdown's selection. For example: select **"Engineering"** → the second dropdown shows only Engineering designations. This requires `INDIRECT` combined with Named Ranges.

01	02	03
Create Named Ranges Create a Named Range for each department's designations — name them "Engineering", "Finance", etc.	First Dropdown Set the first dropdown source to <code>=DeptList</code> to allow department selection.	Second Dropdown Formula Set the second dropdown source to <code>=INDIRECT(A2)</code> where A2 contains the selected department name.

i Build the simple dropdown from the DeptList Named Range first. The dependent dropdown is more complex but demonstrates a real-world form design pattern. If time is short, cover the simple dropdown and mention the dependent one as an advanced technique.

Conditional Formatting: Making Data Visual

Conditional Formatting changes the appearance of cells based on their values — automatically. Research by the Data Warehousing Institute found that analysts using visual formatting identify data anomalies **3x faster** than those working with unformatted tables.

 **Where to find it:** Home tab → Conditional Formatting

Five Built-In Rule Categories

Category	What It Does	Example
Highlight Cell Rules	Colour cells that meet a condition	Salary > ₹10,00,000 → red fill
Top/Bottom Rules	Highlight the highest/lowest values	Top 10% performers → green fill
Data Bars	Add in-cell horizontal bars proportional to value	Salary column gets visual bars
Color Scales	Gradient fill from low to high	Red (low) → yellow (mid) → green (high)
Icon Sets	Small icons based on value thresholds	Traffic lights: green, yellow, red

Advanced Options

New Rule

Create custom formula-based rules — for example, highlight entire rows where Status = "Overdue".

Manage Rules

View, edit, reorder, and delete all rules in the sheet. Essential for maintaining complex formatting logic across large datasets.

Highlight Cell Rules & Top/Bottom Rules

These two rule families are the most frequently applied in professional reporting environments. A 2022 survey of financial analysts found that **74% use Highlight Cell Rules** weekly to flag threshold breaches in KPI dashboards.

Highlight Cell Rules — Condition-Based Colouring

Rule	Highlights Cells Where...
Greater Than	Value > a threshold
Less Than	Value < a threshold
Between	Value is within a range
Equal To	Value matches exactly
Text That Contains	Cell contains specific text
A Date Occurring	Date falls in a relative period
Duplicate Values	Value appears more than once

Top/Bottom Rules — Rank-Based Highlighting

Rule	Highlights...
Top 10 Items	The 10 largest values (customisable)
Top 10%	The top 10% of values (customisable)
Bottom 10 Items	The 10 smallest values
Bottom 10%	The bottom 10% of values
Above Average	All values above the range's average
Below Average	All values below the range's average

Stacking Rules

Multiple conditional formatting rules can be applied to the same range simultaneously. For example: **Top 10% in green** AND **Bottom 10% in red** on the same Salary column. Both rules coexist — top earners glow green, bottom earners glow red, and everyone else remains neutral.



Demo: Apply "Greater Than" on Salary with threshold 1,500,000 (high earners turn red). Then add "Top 10%" in green. Show how both rules layer. Then use "Duplicate Values" on the First_Name column to highlight names appearing more than once.

Data Bars, Color Scales & Icon Sets

Visual encoding transforms raw numbers into instantly interpretable signals. Cognitive science research demonstrates that **pre-attentive visual attributes** — colour, length, and shape — are processed by the human brain up to **250 milliseconds** before conscious attention, making these tools extraordinarily powerful for rapid data comprehension.



Data Bars

Select the Salary column → Conditional Formatting → Data Bars → choose a colour. Each cell gets a horizontal bar proportional to its value. The highest salary gets the longest bar. Great for quick visual comparison without creating a separate chart.



Color Scales

Select Performance_Score → Conditional Formatting → Color Scales. A **2-Color Scale** uses low = red, high = green. A **3-Color Scale** adds yellow in the middle. The gradient makes high and low values instantly obvious across large datasets.



Icon Sets

Select a column → Conditional Formatting → Icon Sets. Options include directional arrows (↑→↓), traffic lights, star ratings, flags, and shapes. Each icon corresponds to a value range. Ideal for dashboards requiring quick status indicators.

Customising Thresholds

After applying any of the above, navigate to **Manage Rules** → **Edit Rule** to change percentage breakpoints, reverse the icon order, or display the icon only (hiding the underlying number). This level of control is essential for professional dashboard design.

- ☐ **Demo:** Apply all three to different columns — Data Bars on Salary, Color Scales on Performance_Score, Icon Sets on Experience_Years. The visual transformation is dramatic: three columns of plain numbers become an instant dashboard. This is the "before and after" moment.

Manage Rules: Control & Priority

When multiple rules apply to the same cells, order matters. In complex reporting workbooks — which enterprise organisations commonly build with **10 or more overlapping formatting rules** — understanding rule priority is essential to producing accurate, predictable visual output.

 **Open Manage Rules:** Home → Conditional Formatting → Manage Rules

What You See in the Dialog

A list of all conditional formatting rules for the current sheet (or selection). Each rule displays: the formula/condition, the format applied, the range it applies to, and its priority order.

Rule Priority Explained

Top-Down Evaluation Rules are evaluated from top to bottom. The first matching rule is applied to each cell.	Stop If True When checked, no further rules are evaluated for that cell — preventing lower-priority rules from overriding.	Conflict Resolution If a cell qualifies for both "Top 10%" (green) and "Greater Than 1,500,000" (red), the rule higher in the list wins .
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Available Actions in Manage Rules

Action	How
Change priority	Select a rule and click the up/down arrows
Edit a rule	Double-click or select + Edit Rule
Delete a rule	Select + Delete Rule
Change the range	Edit the "Applies to" field directly
Stop evaluating	Check "Stop If True" to prevent lower rules from overriding

 **Demo:** Open Manage Rules and display all rules created during this module. Reorder two rules and observe how the cell colour changes. Delete one rule. This demonstrates that conditional formatting requires active management — it is not a "set and forget" feature.

Best Practices: Building Professional Validation & Formatting

Adhering to established best practices is what distinguishes a professionally maintained workbook from an ad hoc spreadsheet. According to KPMG's Data Quality Management framework, **structured input controls and visual governance** are among the top five factors in spreadsheet audit readiness.

1 Always Add Input Messages to Validated Cells

Users should see a helpful tooltip *before* they type, not just an error after they type incorrectly. "Select a department from the list" is far more professional than a cryptic error dialog.

2 Use Named Ranges for Dropdown Sources

=DeptList is maintainable; =\$V\$2:\$V\$11 is fragile. When the list changes, only the name definition needs updating — all dependent dropdowns refresh automatically.

3 Do Not Over-Format

Two or three conditional formatting rules per column is sufficient. Five overlapping rules on the same range creates visual noise and confusion — the opposite of the intended effect.

4 Use Formula-Based Rules for Row-Level Formatting

Highlight Cell Rules only format the evaluated cell. Formula-based rules with a locked column (\$O2) format the entire row — significantly more useful for dashboards and management reports.

5 Review Manage Rules Before Sharing

Hidden or conflicting rules cause confusion for other users. Clean up unused rules and verify priority order before distributing any workbook to stakeholders.

6 Match Visualisation Type to Data Type

Use Color Scales and Data Bars for numeric columns; use Icon Sets for status columns. Mismatched visualisation types reduce clarity rather than enhancing it.

7 Error Alert Types Matter

Stop prevents entry entirely (use for critical validation). **Warning** allows entry after a warning (use for soft guidelines). **Information** merely notifies (use for advisory messages).

Module 8 — Lab Exercise Overview

The following 22-minute lab consolidates all concepts from this module into a single, integrated exercise. Industry benchmarks indicate that **applied practice within 10 minutes of instruction** increases skill retention by up to 75% compared to passive review alone (Learning & Development Institute, 2022).



Part 1 — Data Validation

Apply whole-number validation to Experience_Years (0–50). Add an Input Message and a Stop Error Alert. Test with an out-of-range value.



Part 2 — Dropdown Lists

Build a Department dropdown using the DeptList Named Range. Then construct a cascading Designation dropdown using INDIRECT referencing the selected department.



Part 3 — Highlight Rules

Apply "Greater Than" on Salary (threshold: 1,500,000). Stack a "Top 10%" rule in green. Use "Duplicate Values" on First_Name to identify repeated entries.



Part 4 — Visual Formatting

Apply Data Bars to Salary, a 3-Color Scale to Performance_Score, and Icon Sets to Experience_Years. Observe the dashboard transformation.



Part 5 — Formula-Based Rule

Create a New Rule on range A2:S154 using formula `= $O2="Resigned"` with a light red fill to highlight entire rows for resigned employees.



Part 6 — Manage Rules

Open Manage Rules, review all applied rules, reorder two rules to observe priority effects, and delete any redundant rules before final submission.



Transition note: "You now know how to control input and visualise data. Let's build a complete form with dropdowns, validated fields, and a conditionally formatted dashboard — 10 problems, 22 minutes."

Module 8 — Summary & Next Steps

This module has equipped you with a complete toolkit for **data governance at the cell level**. From preventing erroneous input through validation rules to transforming raw numbers into visual intelligence through conditional formatting, these skills are directly applicable to every professional reporting context.

Data Validation

Restrict input types, ranges, and lists. Add Input Messages and Error Alerts for a professional user experience.

Conditional Formatting

Five built-in categories plus custom formula-based rules for row-level formatting and advanced dashboard design.

1

2

3

4

Dropdown Lists

Direct entry, range references, Named Ranges, and cascading dropdowns using INDIRECT for dynamic form design.

Rule Management

Priority order, Stop If True, conflict resolution, and best practices for maintaining clean, auditable workbooks.

"A spreadsheet that controls its own input and communicates its own insights is not just a tool — it is a self-governing system. That is the standard every professional analyst should aspire to."